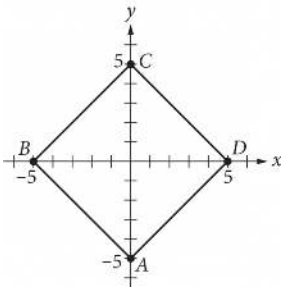
Question ID: cf53cb56

Question



In the xy-plane shown, square ABCD has its diagonals on the x- and y-axes. What is the area, in square units, of the square?

Answer

A. 20

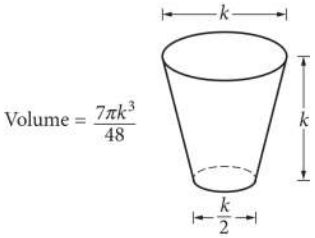
B. 25

C. 50

D. 100

Question ID: 37dde49f

Question



The glass pictured above can hold a maximum volume of 473 cubic centimeters, which is approximately 16 fluid ounces. What is the value of k, in centimeters?

Answer

A. 2.52

B. 7.67

C. 7.79

D. 10.11

Question ID: 468613c0

Question

A triangle has a base length of 56 centimeters and a height of 112 centimeters. What is the area, in square centimeters, of the triangle?

Answer

A. 168

B. $1{,}568$

C. $3{,}136$

D. $6{,}272$

Question ID: a2e76b60

Question

A cylindrical can containing pieces of fruit is filled to the top with syrup before being sealed. The base of the can has an area of $75\ cm^2$, and the height of the can is 10 cm. If $110\ cm^3$ of syrup is needed to fill the can to the top, which of the following is closest to the total volume of the pieces of fruit in the can?

Answer

A. $7.5\ cm^3$

B. $185\ cm^3$

C. $640\ cm^3$

D. $750\ cm^3$

Question ID: aef4fd8a

Question

The length of each side of a square is 94 centimeters (cm). Which expression gives the area, in cm^2, of the square?

Answer

A. $2 \cdot 94$

B. $2 \cdot 94 \cdot 94$

C. $4 \cdot 94$

D. $94 \cdot 94$

Question ID: 9c0a0eca

Question

A triangle has a base length of 10 centimeters and a corresponding height of 70 centimeters. What is the area, in square centimeters, of the triangle?

Answer

A. 700

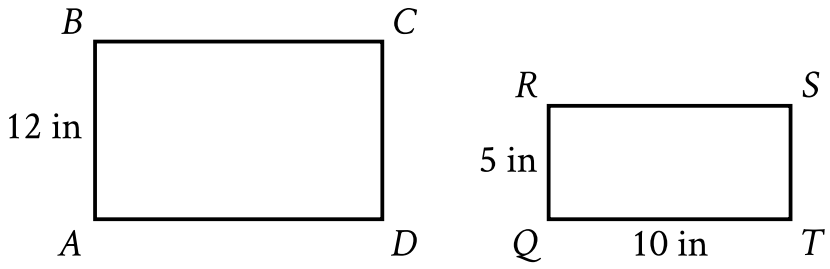
B. 350

C. 175

D. 80

Question ID: e9c5bfb2

Question



Note: Figure not drawn to scale.

Rectangles $ABCD$ and $QRST$ shown are similar, where A, B, C, and D correspond to Q, R, S, and T, respectively. What is the length, in inches (in), of $\overline{AD}$?

Answer

A. 60

B. 24

C. 17

D. 10

Question ID: 151eda3c

Question

A manufacturing company produces two sizes of cylindrical containers that each have a height of 50 centimeters. The radius of container A is 16 centimeters, and the radius of container B is 25% longer than the radius of container A. What is the volume, in cubic centimeters, of container B?

Answer

A. $16{,}000\pi$

B. $20{,}000\pi$

C. $25{,}000\pi$

D. $31{,}250\pi$

Question ID: a2659088

Question

A right circular cylinder has a height of $8\ \mathbf{meters}\ (\mathbf{m})$ and a base with a radius of $12\ \mathbf{m}$. What is the volume, $\mathbf{in}\ \mathbf{m}^3$, of the cylinder?

Answer

A. 8π

B. 20π

C. 768π

D. $1{,}152\pi$

Question ID: 38517165

Question

A circle has a circumference of 31π centimeters. What is the diameter, in centimeters, of the circle?

Question ID: f67e4efc

Question

A right circular cylinder has a volume of 45π. If the height of the cylinder is 5, what is the radius of the cylinder?

Answer

A. 3

B. 4.5

C. 9

D. 40

Question ID: 5afbdc8e

Question

What is the length of one side of a square that has the same area as a circle with radius 2 ?

Answer

A. 2

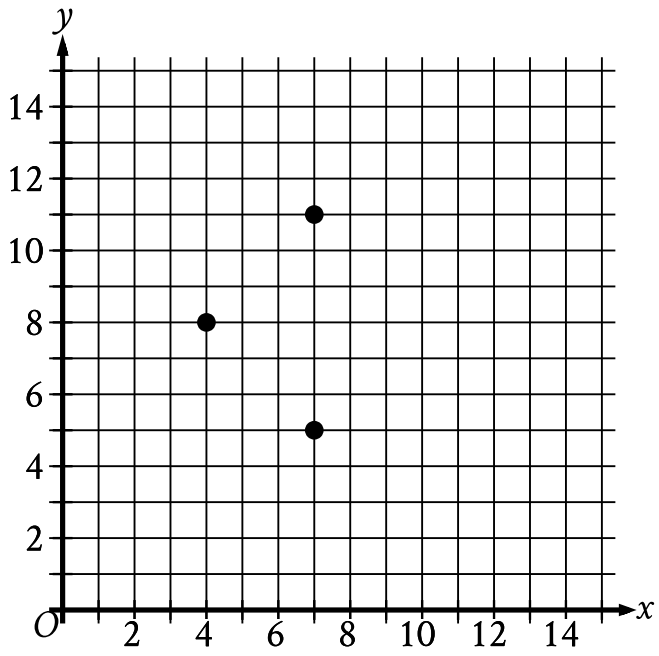
B. $\sqrt{2\pi}$

C. $2\sqrt{\pi}$

D. 2π

Question ID: 096c7ef5

Question



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k?

Answer

A. 3

B. 6

C. 7

D. 9

Question ID: ec5d4823

Question

What is the volume, in cubic centimeters, of a right rectangular prism that has a length of 4 centimeters, a width of 9 centimeters, and a height of 10 centimeters?

Question ID: e336a1d2

Question

A cube has an edge length of 41 inches. What is the volume, in cubic inches, of the cube?

Answer

A. 164

B. $1{,}681$

C. $10{,}086$

D. $68{,}921$

Question ID: c0586eb5

Question

A cylinder has a diameter of 8 inches and a height of 12 inches. What is the volume, in cubic inches, of the cylinder?

Answer

A. 16π

B. 96π

C. 192π

D. 768π

Question ID: 03c6994f

Question

Square A has side lengths that are 246 times the side lengths of square B. The area of square A is k times the area of square B. What is the value of k?

Answer

A. $60{,}516$

B. 492

C. 246

D. 123